

Designing Identity-Affirming Generative AI for Visual Emotional Expression: Empowering Marginalized Young Adults

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Abstract

Technology-supported emotional wellness interventions have gained significant traction in recent years, yet many fail to address the unique needs of marginalized populations whose identities and experiences often go unrecognized in standardized approaches. Building upon prior research documenting how marginality impacts care-seeking behaviors and therapeutic outcomes, this study proposes developing generative AI systems specifically designed to facilitate emotional expression in ways that validate and align with diverse cultural identities and lived experiences. Drawing inspiration from art therapy’s established success in utilizing creative expression for emotional processing, this research deliberately employs design justice principles and highlights “support”-centered design in comparison with traditional emotion regulation approaches. The study examines how generative AI can be tailored to recognize, respect, and respond to the nuanced emotional expressions of individuals from marginalized communities, potentially lowering barriers to mental health support and increasing engagement through personalized, identity-affirming interactions. Through a mixed-methods approach combining interviews, participatory design sessions, and co-design workshops, this research aims to contribute to creating more inclusive AI-assisted emotional support tools that honor the complexity of intersectional identities while providing meaningful pathways to wellness.

Keywords

Mental health, Technology-mediated emotion support, Multimodal Large Language Model, Art Therapy

ACM Reference Format:

Jiaying “Lizzy” Liu. 2025. Designing Identity-Affirming Generative AI for Visual Emotional Expression: Empowering Marginalized Young Adults. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion ’25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3747331>

1 Background & Motivation

We are witnessing an unprecedented convergence of technology and human emotion. Emerging AI-powered chatbots and virtual

“friends” offer round-the-clock companionship and deliver personalized psychological interventions [25]. Meanwhile, other technologies harness immersive virtual reality environments to create safe harbors for those struggling with anxiety [21], and tracking applications foster emotion awareness [9, 11]. Some scholars have found that AI may create harms when mediating humans’ sensitive and affective experiences because these systems are not designed with marginalized populations [26], failing to account for their unique life circumstances and needs [7].

To address these concerns, this work aims to radically re-envision and re-design how we can use emerging AI technologies to authentically serve marginalized communities following human-centered “emotion support” principles [15]. This approach echoes with prior work on cultivating deliberate hope through design to imagine a better future of technologies [10] and AI’s role in human flourishing [23].

In the context of mental health and emotional well-being. Studies found that individuals with limited access to formal healthcare resources—whether due to geographic isolation, financial constraints, or systemic barriers— are more likely to turn to these technologies as substitutes for formal and professional treatment [19]. This suggests opportunities for thoughtfully designed technological interventions that can provide in-situ emotional support [18] when traditional resources remain inaccessible. Visual art-making has long been recognized for its therapeutic potential in emotional expression [3, 20]. Unlike text-restricted chatbots, creative visual expression allows individuals to externalize complex emotions that may be difficult to verbalize [13], particularly when standard terminology doesn’t align with their lived experiences. Art-based approaches benefit emotional wellbeing by providing a non-linear, personalized pathway for processing feelings that transcends linguistic and cultural barriers [8], empowering users to define their emotional landscapes on their own terms.

Despite their potential, deploying general artificial intelligence systems for emotional support among marginalized populations presents significant challenges. Current generative AI models, predominantly trained on datasets reflecting dominant cultural perspectives, often fail to recognize or appropriately respond to culturally-specific emotional expressions and needs [22]. This misalignment creates friction when users from marginalized communities attempt to communicate complex emotional experiences. Generative systems may inadvertently present biases, such as demographic bias [4] that invalidate individuals’ unique lived experiences. It is crucial to calibrate interventions with careful consideration of specific contexts, as failure to do so risks increasing feelings of alienation rather than providing a meaningful connection [15].

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CSCW Companion ’25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3747331>

This study deliberately employs design justice principles [5] to guide the co-imagination of more equitable emotional support technologies, specifically examining how AI image generation systems such as DALL-E and GPT-4o can facilitate drawing-based emotion expression for marginalized young adults. By centering the experiences and perspectives of those traditionally excluded from technology design processes, we explore how generative AI can be configured to represent and resonate with users’ visual self-presentation and emotional realities. This approach creates space for participants to define meaningful visual representations of their emotional states while the system learns to provide responses that create genuine emotional resonance. Through collaborative design sessions with young adults from underrepresented communities, we investigate how generative imagery can serve as a bridge between internal emotional experiences and external validation, potentially creating pathways to emotional processing that honor rather than erase the complex intersections of identity that shape how emotions are experienced and expressed [2].

2 Research Questions & Methodology

Specifically, the research questions of the proposed work include:

- RQ1: What critical design principles should guide technology-mediated emotional support to honor diverse identities and experiences?
- RQ2: How can generative AI be deployed to facilitate emotion expression that cultivates aspiration and agency among marginalized populations?

These questions were built upon our prior work examining how young adults with marginalized identities navigate mental health resources [14]. Our interviews revealed that individuals whose identities exist at the intersection of multiple marginalized dimensions (race, sexuality, gender, socioeconomic status) face unique barriers when seeking mental health and emotional support [16]. The lived experiences of these participants demonstrated how services and technologies that are not sensitive to individuals’ intersectional identities often lead to feelings of alienation rather than support. This identity-centered approach drives our commitment to developing technologies that transcend mere accommodation of difference and instead actively empower individuals by honoring their authentic emotional expressions. We deliberately embrace “aspiration-centered design” [12] to catalyze self-determination and personal growth through the cultivation of identity-affirming emotional connections and the strengthening of individual agency.

As described in Figure 1, I begin with a scoping review that outlines critical design principles for technology-mediated emotion support (Study 1), and will culminate in the participatory creation of identity-affirming AI systems that amplify marginalized voices and experiences (Study 2). This methodological progression ensures that theoretical insights directly inform the development of practical interventions co-created with the communities they aim to serve.

3 Progress& Next Step

Study 1 has been completed, and the next step is to start data collection for Study 2.

3.1 Study 1: Surfacing design principles for technology-mediated emotion interventions (TMEIs)

We conducted a scoping review [15] of 53 empirical studies from ACM proceedings that implemented technology-mediated emotion interventions —digital systems intentionally designed to influence, support, or shape users’ emotional experiences. TMEIs span a range of formats, including chatbot therapists delivering cognitive-behavioral techniques, journaling apps for emotional reflection, mood-sensing wearables that recommend regulation strategies, and gamified platforms that promote gratitude or resilience.

Our analysis revealed a notable intervention dichotomy in the goals of TMEI tools : down-regulate negative emotions or up-regulate positive affect and motivation. Further, we identified approaches beyond narrow adoption of psychological theories around emotion regulation [6] and cognitive behavioral therapy [1]. While many systems draw from psychological models like emotion regulation [6] or cognitive behavioral therapy (CBT) [1], these frameworks are often applied in overly simplistic, “plug-and-play” ways. For instance, CBT-based chatbots designed to replace therapists may overlook safety concerns and cause unintended harms [17, 26]. We argue for more reflexive engagement with these theories, informed by critical perspectives that account for the complex interplay between users, technologies, and their lived contexts.

Situating TMEIs within broader care ecosystems—including patients, caregivers, clinicians, and institutions—our analysis surfaced three dominant roles technologies often play: generating artificial care, intervening in power relationships, and nudging behavioral change. This is especially concerning in care contexts, where individuals face chronic or irreversible conditions and have limited control over their circumstances. In such settings, reductive applications of psychological theories risk invalidating people’s lived emotional realities [24].

In response, we propose a **support-centered emotion intervention** paradigm that embraces emotional complexity rather than prescribing transformation. This approach treats emotions as meaningful and valid responses to uncertainty, rather than problems to be fixed. We identify three design principles to operationalize this perspective:

- validate diverse emotions through *anti-oppressive design principles*
- design rich pathways to embrace (difficult) emotions through *feminist approaches*
- empower people to reclaim agency of emotional narratives through *community-defined well-being practices*

3.2 Study 2: Designing AI-Assisted Visual Emotion Expression Applications for Marginalized Young Adults

Following the findings of Study 1, our Study 2 will employ a two-stage participatory design approach with around 20 young adults (aged between 18 to 24) who self-identify with marginalized identities (including racial/ethnic minorities, LGBTQ+ individuals, and those with intersectional identities).

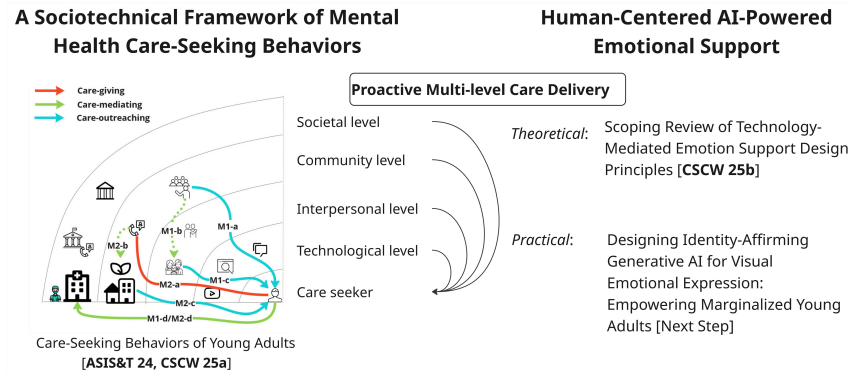


Figure 1: Study Design. The process is described in Section 2.

Stage 1: Individual Co-Design Session. We will conduct in-depth individual co-design sessions (60-90 minutes) with each participant to explore personal experiences of emotion expression and identity representation. These sessions will employ creative elicitation methods where participants create visual artifacts representing their emotional experiences using current generative AI tools, specifically GPT-4o. During these sessions, participants will:

- Generate visual representations of specific emotional states they have experienced (e.g., sadness, loneliness, anger), particularly those connected to their marginalized identities
- Reflect on how well the generated images capture their authentic emotional experiences
- Identify gaps, misrepresentations, or stereotypes in the AI-generated content
- Document which aspects of emotional expression feel culturally or personally misaligned

Participants will evaluate the interaction experience, assessing how the prompt structure, feedback mechanisms, and overall interaction flow either support or hinder authentic emotional expression. Throughout this process, we will investigate whether participants feel a sense of agency and ownership over the visual representations created, or if they experience constraints imposed by the system's limitations that diminish their ability to express their authentic emotional experiences.

All sessions will be audio-recorded and transcribed, with visual artifacts digitally archived (with participant consent) for rigorous analysis. We will document specific prompts used, system responses, and participant modifications to understand the negotiation between human intention and machine interpretation. This comprehensive dataset will inform subsequent design iterations and provide rich qualitative insights into the relationship between identity, emotional expression, and technological mediation.

Stage 2: Evaluation and Testing through Co-design Workshop. Following analysis of the individual sessions, we will synthesize common themes and design directions, developing 3-4 conceptual prototypes that incorporate participants' identity markers into AI-assisted visual emotion expression approaches. Success metrics for these prototypes will be evaluated through:

- Representational Accuracy: Measuring the degree to which participants feel their identity markers are authentically represented (using 1-10 scale ratings and qualitative feedback)
- Emotional Resonance: Assessing how effectively the prototypes convey intended emotions
- Agency and Control: Quantifying participants' perceived level of authorship over the final outputs (using structured interviews and mapping exercises)
- Technical Performance: Tracking reduction in prompt iterations needed to achieve desired results compared to initial sessions

These prototypes will be presented at 3-4 workshops, where groups of participants will evaluate and refine the designs through structured activities. The workshops will include collaborative critique sessions where participants and the researchers will collectively work on improving the prototypes.

4 Contributions

This research aims to contribute to mental health technologies and AI for marginalized populations by centering diverse voices throughout development. We will connect identity markers to AI-powered visual representation systems, establishing guidelines for AI that adapts to varied emotional expressions while preserving agency and cultural integrity.

Our results will inform more responsive AI systems providing identity-affirming emotional support that recognizes cultural specificity, particularly important for young adults who have struggled to find validation within the formal mental health care system. The resulting design guidelines and prototype offer developers pathways to create emotional support technologies that honor diverse self-expression while maintaining user agency, moving toward more equitable interventions that recognize the essential relationship between identity, emotion, and wellbeing.

Acknowledgments

I would like to thank my advisor, Dr. Yan Zhang, for her continued support and thoughtful guidance throughout this journey. I'm also deeply grateful to my collaborators for their generosity and insight, and to the participants in previous projects whose experiences have shaped the direction of this work.

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